

An assignment for



Structuring Conversations into Data

Powered Quality, Root Cause Detection, and CX Insights for FinServeCo

By Bhavya Mahajan

Goal

Deploy an AI Quality Framework to shift FinServeCo's support operations from reactive to proactive by structuring unstructured conversations into data(json) and identifying systemic CX pain points, and delivering real-time insights(dashboard with data points) for Product, Business, and CX teams to Lower repeat contacts, improved satisfaction, and operational efficiency.

Problem Framework: Why AI for Conversation Quality?

TRUE PROBLEM

- FinServeCo’s support, managed by outsourced BPOs, is an operational blind spot. Current metrics (CSAT, NPS, BPO reports) and public signals (Google reviews) fail to capture true quality, masking root causes that drive repeat contacts and agent churn.

EVIDENCE IT’S REAL

- As a B2B fintech, every failed interaction risks client relationships.
- BPO costs are rising without proportional quality gains.
- Lack of rapid root cause detection post-release creates operational risk.

WHY SOLVE NOW

- High repeat contact rates.
- Inconsistent manual tagging vs. actual issue trends and Google reviews.
- Leadership frustration over lack of clarity between product issues vs. agent errors.
- Also, we found out that **72% of customers switch brands** after just one negative experience. ref: [link](#)

WHO FACES THE PROBLEM

- Customers: Templated, low-quality support → frustration & churn.
- CX Managers: No objective data to coach agents → rising operational risk.
- Product/Ops Teams: No visibility into whether issues stem from product bugs or BPO errors.

VALUE GENERATED

Business Impact:

- Acts as an automated compliance & risk audit — flags BPO non-compliance and policy errors.
- Protects revenue by mitigating regulatory exposure and predicting churn early.

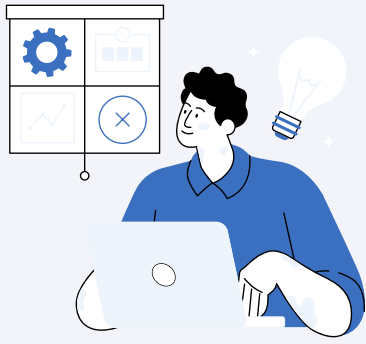
Product Impact:

- Converts unstructured conversations into objective, actionable data.
- Enables root cause-driven product improvements and boosts First Contact Resolution (FCR).

KEY ACTORS



CX Managers



Product Leadership

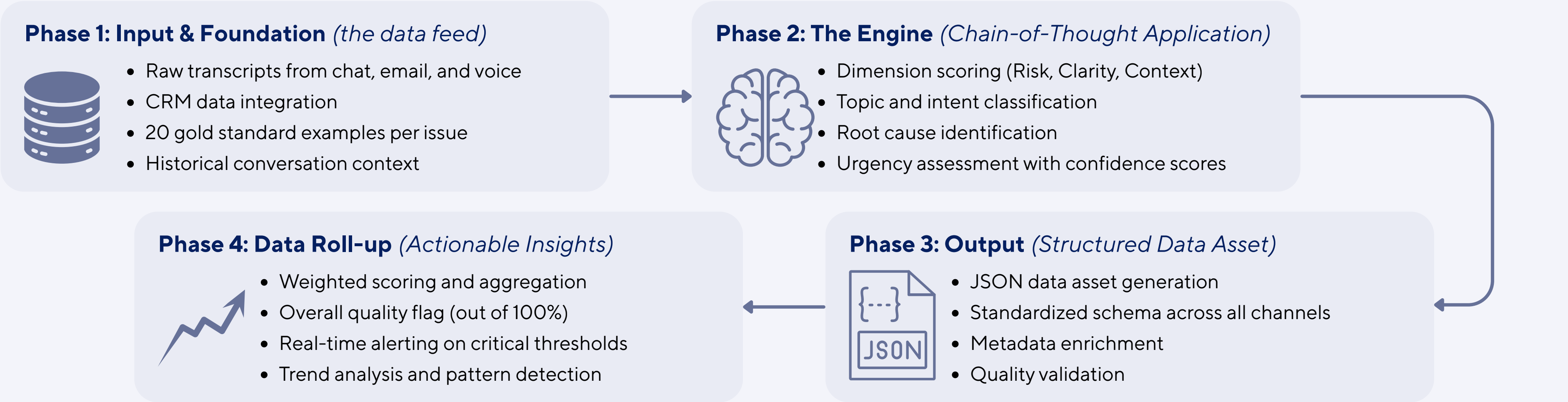


Operations Team



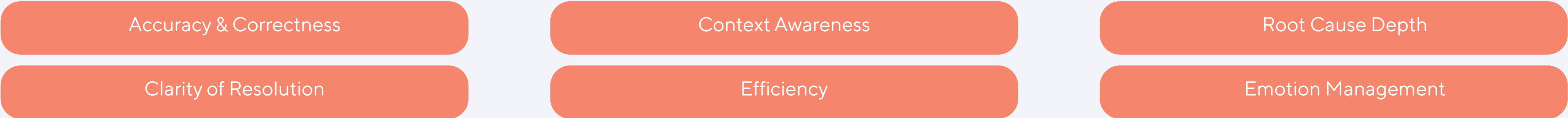
Business Leadership

Pipeline Architecture



Communication Dimensions: Defining Quality

Instead of a single quality score, we evaluate conversations across 6 independent dimensions. Each dimension has specific thresholds that trigger alerts and actions, ensuring the right stakeholder gets notified about the right issue at the right time.



Prompting framework being use: Chain-of-Thought (CoT) prompting

CoT makes the AI think like a quality auditor – structured, transparent, and explainable and breaking down the analysis step-by-step. We would perform analysis on:



AI Prompt Logic for each Dimension

Dimension	AI Prompt Logic	Data Extracted
Accuracy & Correctness Fintech conversations require strict accuracy and compliance.	- Check for mandatory regulatory disclosures (fees, terms, conditions) - Verify accuracy of financial calculations - Detect policy violations or unauthorized promises	- List of disclosures provided vs. required - Accuracy score of financial information given - Security protocol violations (if any)
Use of Context Agents often ignore prior history, leading to repeated troubleshooting.	- Check if agent references prior tickets or conversations - Identify questions that were already answered - Assess proactive use of account/CRM data	- Number of history references made - Count of redundant questions asked - Context utilization percentage
Root Cause Depth Helps separate product vs. process issues.	- Count clarifying questions asked for complex issues - Check if agent captured system logs or error codes - Identify if root cause was documented vs. symptoms	- Root cause identified: Yes/No/Partial - Diagnostic data captured (logs, steps to reproduce) - Number of clarifying questions asked
Clarity of Resolution Vague resolutions cause repeat contacts.	- Verify timeline/expectations are set - Detect contradictory or ambiguous statements - Assess if resolution addresses the original request	- Clarity score: clear next steps provided? - Timeline specified: Yes/No - Confirmation of closure/resolution
Efficiency Long, filler-heavy responses reduce throughput.	- Calculate filler-to-content ratio - Detect redundant phrases and repetition - Assess information density	- Filler percentage (greetings, apologies, fluff) - Response length vs. optimal benchmark - Redundant content detected
Emotion management Impacts trust and churn.	- Detect empathy phrases and acknowledgment of frustration - Check for tone-deaf or dismissive responses - Assess if agent matched emotional tone appropriately	- Starting sentiment score (-1 to +1) - Ending sentiment score (-1 to +1) - Empathy indicators detected

[Click to Redirect : CoT framework prompt ran on a few transcript](#)

How These Dimensions Work Together

The AI generates dimension-specific percentage and flags, which roll up into an overall conversation summary. This multi-dimensional approach ensures:

- **Targeted alerts:** Legal gets compliance issues, CX gets empathy issues, Product gets root cause gaps
- **Nuanced evaluation:** A conversation can be efficient but lack empathy, or vice versa
- **Actionable coaching:** Agents get specific feedback on which skills need improvement
- **Business impact tracking:** Connect dimension percentage and output to outcomes (repeat contacts, CSAT, churn)

JSON Output Schema

Each analyzed conversation produces a structured JSON output that can be stored, queried, and aggregated for insights.

[Click to Redirect : JSON Output Schema](#)

Alert Thresholds & Actions

Dimension	CRITICAL Alert Threshold	Action Triggered
Accuracy & Correctness	< 70% (Failure to follow >= 2\$ major policies, or a a 100% security failure).	Legal/Compliance Alert. Ticket must be immediately reviewed and potentially reported. Agent faces disciplinary action.
Use of Context	< 40% (Agent fails to check history and forces the customer to repeat themselves entirely).	CX Manager Coaching Alert. Agent requires immediate training on using the CRM/history features.
Root Cause Depth	< 50% (Agent immediately escalated a known issue without capturing required diagnostic data).	Product/SME Alert. The agent is failing to gather critical data needed for the Product team to fix the underlying systemic bug.
Clarity of Resolution	< 50% (The resolution is ambiguous, factually contradictory, or has no clear next step).	Operations/Process Alert. The conversation is highly likely to generate a Repeat Contact within 72 hours.
Efficiency	< 30% (Response contained over {70\%} filler/redundant content, wasting significant time).	CX Manager Coaching. The agent is costing the business money by taking too long and reducing overall throughput.
Emotion management	0% (Customer sentiment ends up more negative than the starting sentiment).	CX Manager Review. The conversation is flagged as high risk for client attrition/churn.

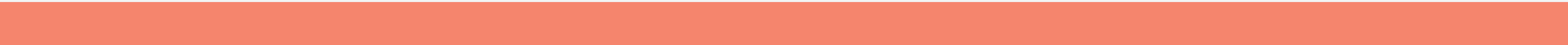
Insight Philosophy: From Data to Decision

Each insight type is designed with a specific stakeholder and decision in mind. We don't just report what happened—we surface why it matters and what action to take. Insights are delivered through the right channel at the right time: real-time alerts for critical issues, daily dashboards for trends, weekly digests for strategic planning.



Insight Philosophy: From Data to Decision

Insights	Frequency	Channel	Audience	Visuals
Post-Release Spikes	Real-time ▾	Slack alert	Product, Ops	- Slack alert: Real-time critical threshold breach. - Operations Dashboard: Spike timeline with release markers.
Friction Themes	Weekly ▾	Email + Dashboard	CX Manager	- Dashboard: Bubble chart sized by volume, colored by severity, with trend arrows. - Weekly digest email to Product.
Knowledge Gap	Weekly ▾	Dashboard + Email	Product, CX	- Opportunity matrix (volume vs. complexity). - Weekly summary with top 5 self-service candidates.
Root Cause Impact	Monthly ▾	Dashboard + Email	Leadership	- Product Dashboard: Ranked table with impact scores, cost estimates, and trend indicators. - Monthly executive summary.
Empathy Trends	Daily ▾	Dashboard + Slack alert	CX, Product	- CX Dashboard: Heatmap by agent, line chart over time. - Slack alert when any agent drops below threshold.



Insight Philosophy: From Data to Decision

Insights	How AI Detects This Insight	Why It Matters	Decision Enabled
Post-Release Spikes	Time-series anomaly detection + Topic correlation - Establish baseline ticket volume and sentiment per topic - Correlate spikes with product release calendar and feature flags - Auto-classify spike severity based on magnitude and acceleration	Catch critical bugs or UX issues within hours of release, not weeks	Rollback feature vs. communicate known issue vs. emergency patch
Friction Themes	Semantic similarity + Topic modeling - Use embedding-based clustering to group similar issues - Rank clusters by frequency, severity, and trend velocity	Product teams can prioritize roadmap based on actual user friction rather than anecdotal feedback	Allocate engineering resources to fix authentication bugs vs. improving onboarding flow
Knowledge Gap	FAQ clustering + Resolution pattern analysis - Group tickets by customer intent and question type - Identify repetitive, low-complexity queries with consistent solutions - Suggest specific help articles, chatbot flows, or UI tooltips	Reduce support volume through self-service, freeing agents for complex issues	Create help article vs. add in-app tooltip vs. build chatbot flow
Root Cause Impact	Causal inference + Aggregation - Tag each ticket with root cause using classification model - Track subsequent actions: refund issued, escalation, account cancellation - Calculate impact metrics per root cause: avg resolution time, cost, retention risk,cluster urgency	Not all issues are equal - prioritize fixes that reduce refunds, escalations, and churn risk	Focus on "billing duplicate charge" issue (high refund cost) vs. "slow dashboard load" (low impact)
Empathy Trends	Sentiment analysis + Linguistic pattern matching - Analyze empathy_tone dimension across all agent conversations - Calculate baseline empathy scores per agent and team - Detect statistically significant drops week-over-week	Proactive coaching prevents CSAT degradation and agent burnout before it impacts customers	Identify which agents need empathy training vs. which are modeling best practices

[Click to Redirect: Dashboard View](#)

Operational Planning and Communication: 3-month pilot roadmap with clear milestones, deliverables, and stakeholder communication

[Click to Redirect: Product Roadmap](#)

Communication Rhythm

Daily – Slack Standup:

Audience: Core Team

Purpose: Track blockers and daily progress

Weekly – Email + Dashboard Snapshot:

Audience: CX, Product, Ops leads

Monthly – Recorded Demo & Written Summary:

Audience: Broader Organization

Communication Examples:

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Next Daily Stand-up (Slack Summary – Week 4 Day 2)

Yesterday:

- ✓ Completed end-to-end test of scoring engine (CoT → JSON → Summary view)
- ✓ Verified accuracy on 500 tickets = 82% vs human baseline
- ✓ Legal review in progress for critical-alert wording

Today:

-  CX Team – Review Agent Coaching dashboard wireframe (Context & Empathy focus)
-  Product Team – Finalize Root Cause dashboard data structure
-  Ops Team – Begin alert-threshold tuning (Risk & Accuracy dimension)

Blockers / Needs:

-  Waiting for BPO-2 sample upload (1 K tickets)
-  Need sign-off from Legal on compliance-alert messaging

Note: On track for **Week 5 milestone – Dashboard Alpha Build** by Friday.

 Aa     

To: cx-team@finserveco.com product-leads@finserveco.com ops-team@finserveco.com

Cc:

Subject: FinServeCo AI Quality Framework Pilot – Week 3 Update

Hi Team,

As we close Week 3 of the AI Quality Framework Pilot, we've successfully completed the Foundation & Calibration Phase and are now moving into Model Development & Validation (Phase 2).

Pilot Highlights

1. All core setup, data access, and baseline calibration activities are now complete.
2. Data Access & Pipeline Setup Complete
3. Gold Standard Label Set Created
4. Prompt Library v1.0 Released

The first version of the AI prompt framework for 5 communication dimensions is live:
Use of Context, Accuracy & Correctness, Root Cause Depth, Clarity of Resolution, and Emotion Management.

Pilot Results

Baseline Metrics Captured
Initial benchmark shows:
Avg CSAT = 74%
Repeat Contact Rate = 31%
Average Conversation Length = 4.8 messages
These serve as our pre-AI baselines for later impact measurement.

Next Steps:

- Begin model training and validation using 30 days of historical data
- Test topic classification and intent detection models
- Start defining root cause and urgency scoring logic
- Align with Legal & Ops on alert threshold parameters

Blockers / Dependencies

- Awaiting IT confirmation on full CRM export automation (expected early next week)
- Need CX Team input on empathy dimension examples for fine-tuning

Thank you for helping us move quickly through setup – this strong foundation ensures smoother validation ahead.

Best,
Bhavya Mahajan
Product Specialist – AI
SentiSum

Evaluation Metrics & Validation of Prompt framework

Validation ensures that the model's scores are accurate, explainable, and aligned with FinServeCo's communication quality standards.

We use a **two-layer validation process**: Human-in-the-Chain (for grounding) and LLM-as-Evaluator (for scale).

A. Human-in-the-Chain Validation (Expert Ground Truth)

- Human evaluators establish the gold standard before the AI is assessed by adding labels to 20-30 conversations. This ensures that FinServeCo's unique tone, policy, and empathy standards are captured.
- This becomes the ground truth reference for the next stage.

B. LLM-as-Evaluator Validation

- Once human agreement is achieved, we test how well the AI replicates expert reasoning.
- The AI's outputs are compared directly against the human majority labels.
 - **G- eval (Reference- Based Evaluation)**
 - Calculate agreement rate (target $\geq 85\%$)
 - Measure Recall and Precision per dimension
 - For policy violations (critical alert):
 - Prioritize RECALL - Accept some false positives for human review
 - For empathy coaching (non-critical):
 - Use F1-score for optimization
 - Identify Systemic errors for prompt Refinement

[Click to Redirect: AI Framework Performance metrics](#)

